FA 5: Palmer Penguins

2026-10-11

Part A. Data Preparation

```
# Download and import the dataset
dir.create("data", showWarnings = FALSE)

url <- "https://raw.githubusercontent.com/allisonhorst/palmerpenguins/main/inst/extdata/penguins.csv"

if (!file.exists("data/penguins.csv")) {
  download.file(url, "data/penguins.csv", mode = "wb")
}

penguins <- read.csv("data/penguins.csv")

# Examine the dataset
str(penguins)
```

```
## 'data.frame':    344 obs. of  8 variables:
##  $ species          : chr  "Adelie" "Adelie" "Adelie" "Adelie" ...
##  $ island           : chr  "Torgersen" "Torgersen" "Torgersen" "Torgersen" ...
##  $ bill_length_mm   : num  39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42 ...
##  $ bill_depth_mm    : num  18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2 ...
##  $ flipper_length_mm: int  181 186 195 NA 193 190 181 195 193 190 ...
##  $ body_mass_g      : int  3750 3800 3250 NA 3450 3650 3625 4675 3475 4250 ...
##  $ sex              : chr  "male" "female" "female" NA ...
##  $ year             : int  2007 2007 2007 2007 2007 2007 2007 2007 2007 2007 ...
```

```
summary(penguins)
```

```
##    species             island          bill_length_mm  bill_depth_mm
##  Length:344         Length:344         Min.   :32.10   Min.   :13.10
##  Class :character   Class :character   1st Qu.:39.23   1st Qu.:15.60
##  Mode  :character   Mode  :character   Median :44.45   Median :17.30
##                                        Mean   :43.92   Mean   :17.15
##                                        3rd Qu.:48.50   3rd Qu.:18.70
##                                        Max.   :59.60   Max.   :21.50
##                                        NA's   :2       NA's   :2
##  flipper_length_mm  body_mass_g       sex                year
##  Min.   :172.0     Min.   :2700   Length:344         Min.   :2007
##  1st Qu.:190.0     1st Qu.:3550   Class :character   1st Qu.:2007
##  Median :197.0     Median :4050   Mode  :character   Median :2008
##  Mean   :200.9     Mean   :4202                      Mean   :2008
##  3rd Qu.:213.0     3rd Qu.:4750                      3rd Qu.:2009
##  Max.   :231.0     Max.   :6300                      Max.   :2009
##  NA's   :2         NA's   :2
```

```r
# Identify missing values
colSums(is.na(penguins))
```

```
##           species            island    bill_length_mm     bill_depth_mm
##                 0                 0                 2                 2
## flipper_length_mm       body_mass_g               sex              year
##                 2                 2                11                 0
```

```r
# Remove rows missing the required variables
penguins_clean <- subset(
  penguins,
  !is.na(body_mass_g) &
    !is.na(species) &
    !is.na(sex)
)

# Convert variables into factors
penguins_clean$species <- factor(
  penguins_clean$species,
  levels = c("Adelie", "Chinstrap", "Gentoo")
)

penguins_clean$sex <- factor(
  penguins_clean$sex,
  levels = c("female", "male")
)

# Final sample size
nrow(penguins_clean)
```

```
## [1] 333
```

Dataset: Palmer Penguins

Response variable: Body mass in grams

Categorical variables: Species and sex

Missing observations: The original dataset has 344 rows and missing values in several columns.

Final sample size: $n = 333$

Part B. Exploratory Data Analysis

```r
# Overall descriptive statistics
x <- penguins_clean$body_mass_g

mean(x)
```

```
## [1] 4207.057
```

```r
sd(x)
```

```
## [1] 805.2158
```

```r
min(x)
```

```
## [1] 2700
```

```r
max(x)
```

```
## [1] 6300
```

```r
# Statistics by species
aggregate(
  body_mass_g ~ species,
  data = penguins_clean,
  FUN = function(x) c(
    n = length(x),
    mean = mean(x),
    sd = sd(x)
  )
)
```

```
##     species body_mass_g.n body_mass_g.mean body_mass_g.sd
## 1    Adelie      146.0000        3706.1644       458.6201
## 2 Chinstrap       68.0000        3733.0882       384.3351
## 3    Gentoo      119.0000        5092.4370       501.4762
```

```r
# Statistics by sex
aggregate(
  body_mass_g ~ sex,
  data = penguins_clean,
  FUN = function(x) c(
    n = length(x),
    mean = mean(x),
    sd = sd(x)
  )
)
```

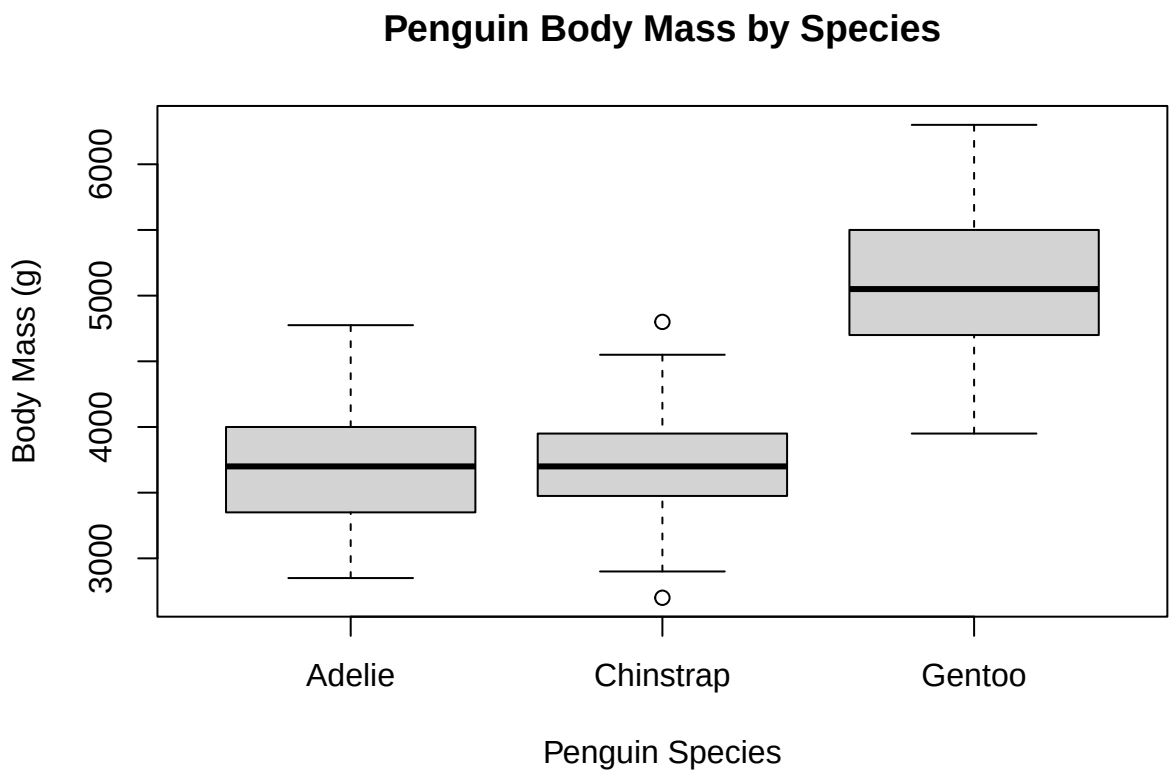
```
##      sex body_mass_g.n body_mass_g.mean body_mass_g.sd
## 1 female      165.0000        3862.2727       666.1720
## 2   male      168.0000        4545.6845       787.6289
```

```r
# Statistics by species and sex
aggregate(
  body_mass_g ~ species + sex,
  data = penguins_clean,
  FUN = function(x) c(
    n = length(x),
    mean = mean(x),
    sd = sd(x)
  )
)
```

```
##     species    sex body_mass_g.n body_mass_g.mean body_mass_g.sd
## 1    Adelie female       73.0000        3368.8356       269.3801
## 2 Chinstrap female       34.0000        3527.2059       285.3339
## 3    Gentoo female       58.0000        4679.7414       281.5783
## 4    Adelie   male       73.0000        4043.4932       346.8116
## 5 Chinstrap   male       34.0000        3938.9706       362.1376
## 6    Gentoo   male       61.0000        5484.8361       313.1586
```

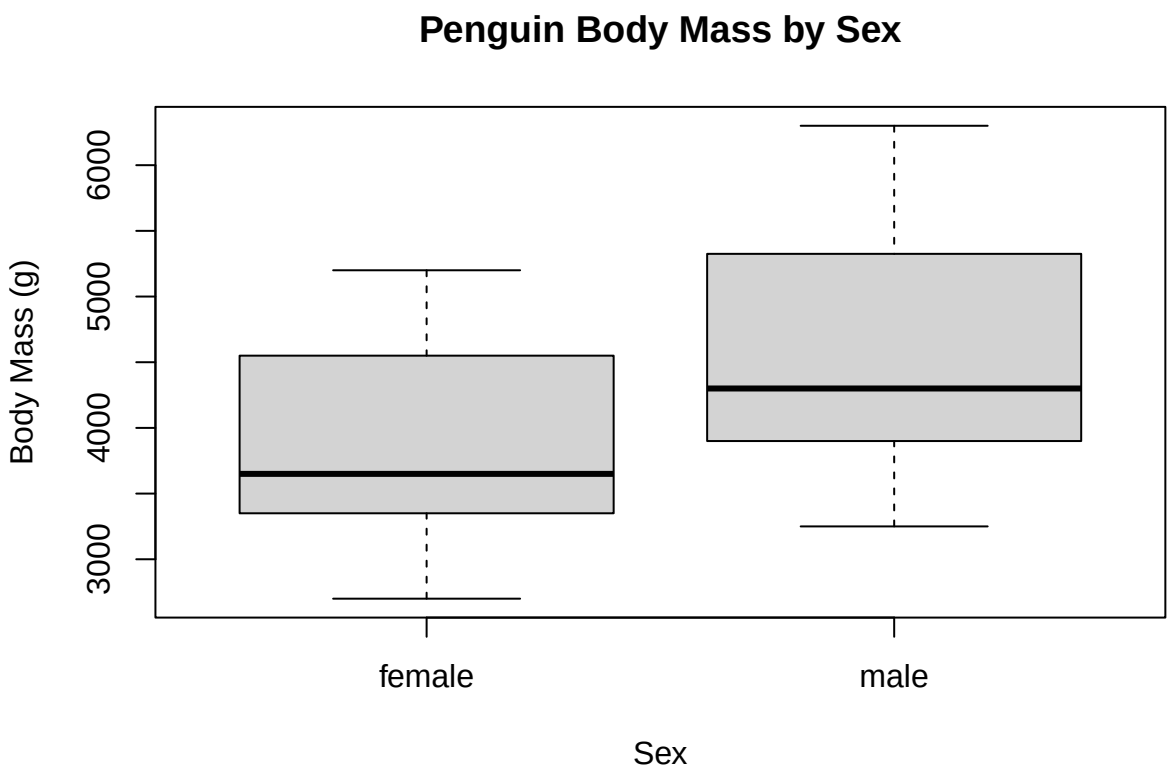
Visualization

```r
# Boxplot by species
boxplot(
  body_mass_g ~ species,
  data = penguins_clean,
  xlab = "Penguin Species",
  ylab = "Body Mass (g)",
  main = "Penguin Body Mass by Species"
)
```




```r
# Boxplot by sex
boxplot(
  body_mass_g ~ sex,
  data = penguins_clean,
  xlab = "Sex",
  ylab = "Body Mass (g)",
```

```
  main = "Penguin Body Mass by Sex"
)
```



Describe the Patterns

Which species has the highest average body mass? Gentoo penguins.

Which sex has the higher average body mass? Male penguins.

Are the male–female differences similar across species? No. The differences vary by species. The male–female difference is about 675 g for Adelie, 412 g for Chinstrap, and 805 g for Gentoo penguins.

Part C. One-Way ANOVA

```
oneway_model <- aov(
  body_mass_g ~ species,
  data = penguins_clean
)

summary(oneway_model)
```

```
##              Df    Sum Sq  Mean Sq F value Pr(>F)
## species       2 145190219 72595110   341.9 <2e-16 ***
## Residuals   330  70069447   212332
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

a. Null hypothesis The mean body mass is equal for all three species.

$$H_0 : \mu_A = \mu_C = \mu_G$$

b. Alternative hypothesis At least one species has a different mean body mass.

c. ANOVA table

Source	df	Sum Sq	Mean Sq	F-value	p-value
Species	2	145,190,219	72,595,110	341.9	$< 2 \times 10^{-16}$
Residuals	330	70,069,447	212,332	—	—

d. F-statistic, degrees of freedom, and p-value

F-statistic: approximately 341.9

Degrees of freedom: $(2, 330)$

p-value: $p < 0.001$

e. Statistical decision

Reject the null hypothesis because the p-value is below 0.05.

f. Interpretation

There is a statistically significant difference in mean body mass among penguin species. At least one species has a different mean body mass from another.

Part D. Dummy-Variable Regression

```
# Adelie is the reference category
dummy_model <- lm(
  body_mass_g ~ species,
  data = penguins_clean
)

summary(dummy_model)
```

```
##
## Call:
## lm(formula = body_mass_g ~ species, data = penguins_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1142.44  -342.44   -33.09   307.56  1207.56
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)       3706.16      38.14  97.184   <2e-16 ***
## speciesChinstrap    26.92      67.65   0.398    0.691
## speciesGentoo     1386.27      56.91  24.359   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 460.8 on 330 degrees of freedom
## Multiple R-squared:  0.6745, Adjusted R-squared:  0.6725
## F-statistic: 341.9 on 2 and 330 DF,  p-value: < 2.2e-16
```

a. Reference category: Adelie

b. Dummy variables:

- speciesChinstrap: 1 for Chinstrap, 0 otherwise.
- speciesGentoo: 1 for Gentoo, 0 otherwise.

c. Regression coefficients

The approximate model is:

$$\hat{Y} = 3700.66 + 32.43D_C + 1375.35D_G$$

Where:

$\hat{Y}$ = predicted body mass in grams.

D_C = Chinstrap dummy variable.

D_G = Gentoo dummy variable.

The Chinstrap-versus-Adelie difference is not statistically significant ($p \approx 0.881$); the Gentoo-versus-Adelie difference is highly significant ($p < 0.001$).

d. Relationship between dummy regression and One-Way ANOVA Both methods examine differences in mean body mass among species. The One-Way ANOVA tests whether any species means differ, while dummy-variable regression estimates the differences relative to the reference species, Adelie.

Part E – Two-Way ANOVA

```
twoway_model <- aov(
  body_mass_g ~ species * sex,
  data = penguins_clean
)

summary(twoway_model)
```

```
##              Df    Sum Sq  Mean Sq F value   Pr(>F)
## species       2 145190219 72595110 758.358  < 2e-16 ***
## sex           1  37090262 37090262 387.460  < 2e-16 ***
## species:sex   2   1676557   838278   8.757 0.000197 ***
## Residuals   327  31302628    95727
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Species: H_0: Mean body mass is equal across species.

Sex: H_0: Mean body mass is equal between females and males.

Interaction: H_0: The effect of species on mean body mass does not depend on sex.

Source	Sum Sq	df	Mean Sq	F-value	p-value
Species	145,190,219	2	72,595,110	758.358	$< 2 \times 10^{-16}$
Sex	37,090,262	1	37,090,262	387.460	$< 2 \times 10^{-16}$
Species × Sex	1,676,557	2	838,278	8.757	0.000197
Residuals	31,302,628	327	95,727	—	—

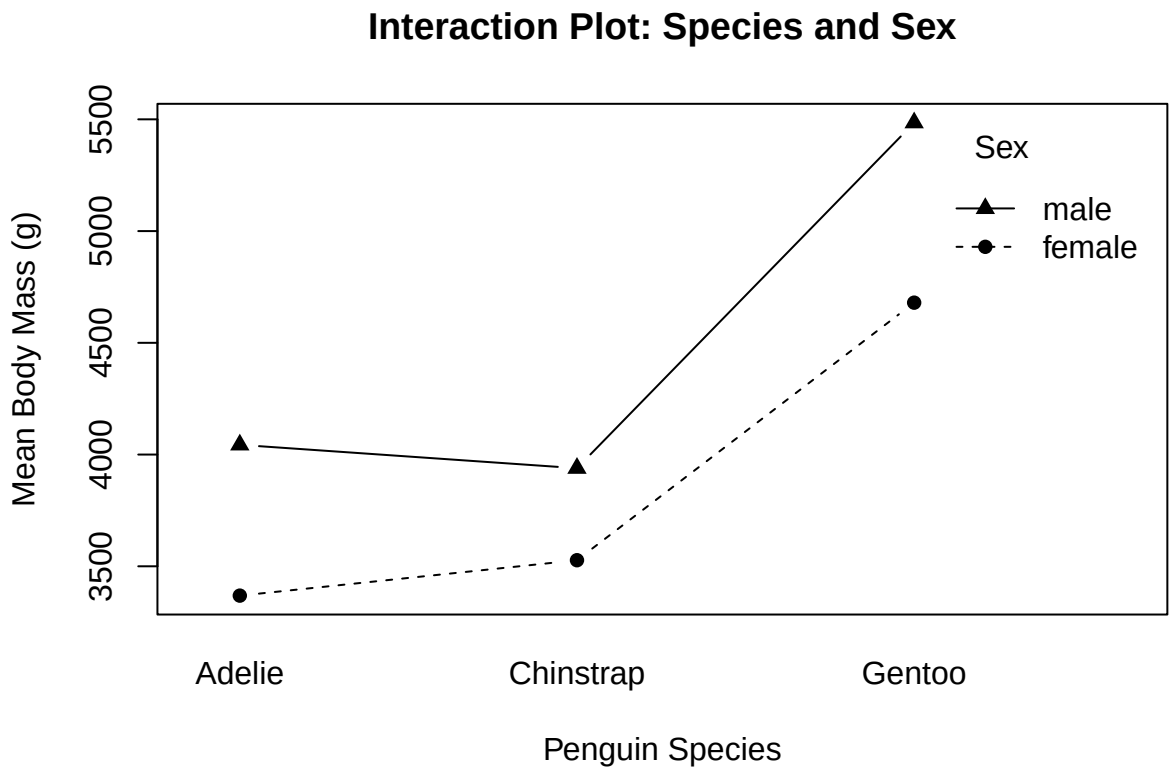
Interpretation of each effect

- ***Species:*** Significant. Mean body mass differs among species after accounting for sex in this model.
- ***Sex:*** Significant. Male and female body mass differ.
- ***Species × Sex:*** Significant. The male–female difference changes across species.

Decision: Reject all three null hypotheses at the 0.05 significance level.

Part F. Interaction Analysis

```
interaction.plot(
  x.factor = penguins_clean$species,
  trace.factor = penguins_clean$sex,
  response = penguins_clean$body_mass_g,
  fun = mean,
  type = "b",
  pch = c(16, 17),
  xlab = "Penguin Species",
  ylab = "Mean Body Mass (g)",
  trace.label = "Sex",
  main = "Interaction Plot: Species and Sex"
)
```



a. Are the profiles approximately parallel? No. The lines are not perfectly parallel.

b. If they are nonparallel, what does this suggest? It suggests a possible interaction between species and sex.

c. Is the Species × Sex interaction statistically significant? Yes. The interaction has $F(2, 327) \approx 8.757$ and $p \approx 0.000197$, which is below 0.05.

d. Does the difference between male and female body mass appear to change across species? Yes. The difference is approximately 675 g for Adelie, 412 g for Chinstrap, and 805 g for Gentoo penguins.

e. Explain the interaction in practical terms. The difference in body mass between male and female penguins depends on their species. Although males have higher average body mass in all three species, the size of the difference is not the same for each species.

Part G. Full Regression Model

```
full_model <- lm(
  body_mass_g ~ species * sex,
  data = penguins_clean
)

summary(full_model)
```

```
##
## Call:
## lm(formula = body_mass_g ~ species * sex, data = penguins_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -827.21 -213.97   11.03  206.51  861.03
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)                3368.84      36.21  93.030  < 2e-16 ***
## speciesChinstrap            158.37      64.24   2.465  0.01420 *
## speciesGentoo              1310.91      54.42  24.088  < 2e-16 ***
## sexmale                     674.66      51.21  13.174  < 2e-16 ***
## speciesChinstrap:sexmale   -262.89      90.85  -2.894  0.00406 **
## speciesGentoo:sexmale       130.44      76.44   1.706  0.08886 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 309.4 on 327 degrees of freedom
## Multiple R-squared:  0.8546, Adjusted R-squared:  0.8524
## F-statistic: 384.3 on 5 and 327 DF,  p-value: < 2.2e-16
```

```
anova(full_model)
```

```
## Analysis of Variance Table
##
## Response: body_mass_g
##              Df    Sum Sq  Mean Sq F value    Pr(>F)
## species       2 145190219 72595110 758.358 < 2.2e-16 ***
## sex           1  37090262 37090262 387.460 < 2.2e-16 ***
## species:sex   2   1676557   838278   8.757 0.0001973 ***
## Residuals   327  31302628    95727
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- ***Intercept***

  The intercept is approximately 3,368.84 g. It represents the predicted mean body mass of female Adelie penguins.

- ***Species coefficient(s)***

  Chinstrap coefficient: approximately +158.37 g.

  Gentoo coefficient: approximately $+1,310.91$ g.

- ***Sex coefficient***

  The male coefficient is approximately +674.66 g. For Adelie penguins, males have a predicted mean body mass about 675 g higher than females.

- ***Species × Sex interaction coefficient(s)***

  Chinstrap × male: approximately −262.89 g.

  Gentoo × male: approximately +130.44 g.

These coefficients show how the male–female difference for each species changes relative to the difference for Adelie penguins.

Final Interpretation

The analysis shows that penguin body mass differs significantly according to species and sex. Gentoo penguins have the highest average body mass, and males are heavier on average than females in all three species. The significant Species × Sex interaction indicates that the difference between male and female body mass varies by species. Therefore, both species and sex are important factors in understanding penguin body mass, and their relationship should be interpreted together.